CVLB2301T-Concrete Technology

(Session 2026–27)
SEMESTER III

Subject Code:CVLB2301T Credits:4.0
Time: 3 Hrs**Max. Marks:** 100**Min Pass Marks:** 40%**External:** 50

(A) COURSE OBJECTIVES

At the completion of the course, the students will be able to:

- Understand the properties and behavior of concrete materials such as cement, aggregates, and water.
- Gain knowledge of fresh concrete properties including workability, segregation, and bleeding.
- Understand hardened concrete properties such as strength, durability, and stress-strain behavior.
- Learn mix design procedures using standard methods (ACI and IS codes).
- Understand the use of admixtures, waste materials, and modern concreting techniques.
- Analyze causes of deterioration and apply repair and rehabilitation techniques in concrete structures.

(B) SYLLABUS

SECTION A

Concrete Materials:
Introduction to concrete materials – Cement: physical tests – Aggregates: tests – Quality of water for mixing and curing – Use of sea water in concrete.

Fresh and Hardened Concrete:
Workability – Tests for workability – Segregation and bleeding – Strength properties of hardened concrete – Compressive strength – Split tensile strength – Flexural strength – Stress-strain curve – Modulus of elasticity – Durability – Water absorption – Permeability – Corrosion test – Acid resistance.

Admixtures and Waste Materials:
Accelerating, retarding, water-reducing and air-entraining admixtures Coloring agents – Plasticizers – Use of waste materials in concrete.

Concrete Production:
Batching – Mixing – Transportation – Placing – Curing of concrete.

Mix Design:
Factors influencing mix proportion – Mix design by ACI method and IS code method.

SECTION B

Concrete Behavior:
Strength of concrete – Shrinkage and temperature effects – Creep – Corrosion: causes, effects, and remedial measures – Thermal properties.

Cracking and Repair:
Micro-cracking – Symptoms – Evaluation of cracks – Types and methods of crack repair.

Special Concrete and Techniques:
Lightweight concrete – Fiber reinforced concrete – Polymer and polymer-modified concrete – Ready mix concrete – Self-compacting concrete – Pumped concrete – Underwater concrete – Pre-placed concrete – Vacuum dewatered concrete – Hot and cold weather concreting.

Repair and Rehabilitation:
Distress in structures – Causes and precautions – Damage assessment – Repairing techniques and materials.

Prestressed Concrete (Introduction):
Basic concepts – Classification and types – Prestressing systems – Pretensioned and post-tensioned concrete elements. *(No numericals)*

(C) RECOMMENDED BOOKS

1. Shetty, M.S., Concrete Technology: Theory & Practice, S. Chand & Co., 2004.
2. Gambhir, M.L., Concrete Technology, Tata McGraw Hill, 2004.
3. Neville, A.M., Properties of Concrete, Longman Publishers, 2004.
4. Santhakumar, A.R., Concrete Technology, Oxford University Press, New Delhi, 2007.

➢ **Course Profile**

Course Title	Concrete Technology
Type	Core Course (UG Programme)
Session / Semester	2026–27 / Semester III
Subject Code	**CVLB2301T**
Credits	4
L–T–P	3–1–0
Maximum Marks	**100**
Evaluation Scheme	External (50) + Internal (50)
End-Sem Exam Duration	3 Hours
Minimum Passing Marks (External)	40%

CVLB2302T-Survey & Geomatics

(Session 2026–27)
SEMESTER III

Subject Code:CVLB2302T Credits:4.0
Time: 3 Hrs**Max. Marks:** 100**Min Pass Marks:** 40%**External:** 50

(A) COURSE OBJECTIVES

At the completion of the course, the students will be able to:

- Understand basic concepts and types of surveying.
- Learn the working principles and applications of surveying instruments.
- Gain knowledge of chain, compass, and plane table surveying methods.
- Understand leveling and contouring techniques for elevation measurement.
- Develop skills to measure distances, angles, and elevations accurately.
- Apply surveying techniques for area and volume calculations.

(B) SYLLABUS

SECTION A

Introduction:
Different types of surveys.

Chain Surveying:
Principle of chain surveying – Description of equipment – Methods of chaining and booking – Selection of base line and stations – Obstacles in chaining – Location of inaccessible points using chain, tape, and ranging rods.

Prismatic Compass Survey:
Description of prismatic and surveyor's compass – Methods of traversing – Local attraction and its elimination – Adjustment of closing error by graphical method.

SECTION B

Plane Table Survey:
Description of equipment – Methods of plane tabling – Strength of fix – Two-point and three-point problems and their solutions.

Leveling:
Dumpy and tilting levels – Leveling staves – Methods of leveling – Sensitivity of bubble tube – Setting out grade lines – Permanent adjustment of leveling instruments.

Contouring:
Contour gradient – Methods of contouring – Earthwork calculations of areas and volumes.

Minor Instruments:
Box sextant – Hand level – Abney level – Planimeter – Ghat tracer – Tangent clinometer, etc.

(C) RECOMMENDED BOOKS

1. Kanetkar, T.P., *Surveying Vol. I & II*, Pune Vidyarthi Griha Prakashan (1985).
2. Sahiwney, P.B., *Surveying*.
3. Singh, Narinder, *Surveying*, Tata McGraw Hill (1992).
4. Punmia, B.C., *Surveying Vol. I & II*, Laxmi Publications (1998).
5. Agor, R., *Surveying*, Khanna Publishers (1982).
6. Venkataramiah, C., *A Text Book of Surveying*, Universities Press (1996).

➢ **Course Profile**

Course Title	Survey & Geomatics
Type	Minor Course (UG Programme)
Session / Semester	2026–27 / Semester III
Subject Code	**CVLB2302T**
Credits	4.0
L–T–P	3–1–0
Maximum Marks	100
Evaluation Scheme	External (50) + Internal (50)
End-Sem Exam Duration	3 Hours
Minimum Passing Marks (External)	40%

CVLB2303T-FLUID MECHANICS

(Session 2026–27)
SEMESTER III

Subject Code:CVLB2303TCredits:4.0
Time: 3 Hrs**Max. Marks:** 100**Min Pass Marks:** 40%**External:** 50

(A) COURSE OBJECTIVES

At the completion of the course, the students will be able to:

- Understand fundamental properties and behavior of fluids.
- Apply principles of fluid statics to analyze forces on submerged surfaces and floating bodies.
- Understand fluid kinematics and describe fluid motion using various approaches.
- Apply principles of fluid dynamics including Bernoulli's equation to solve engineering problems.
- Analyze flow through pipes including losses and energy considerations.
- Understand dimensional analysis and similitude for model studies.
- Measure flow using different devices and apply concepts in practical problems.

(B) SYLLABUS

SECTION A

Fluid Properties:
Concept of fluid – Difference between solids, liquids, and gases – Ideal and real fluids – Continuum concept – Density, specific weight, relative density – Viscosity and temperature dependence – Surface tension and capillarity – Vapour pressure and cavitation – Compressibility and bulk modulus – Newtonian and non-Newtonian fluids.

Fluid Statics:
Pressure and Pascal's law – Hydrostatic paradox – Forces on plane (horizontal, vertical, inclined) and curved surfaces – Center of pressure – Buoyancy and floatation – Stability of floating and submerged bodies – Metacentric height – Rotation of liquid in cylindrical containers.

Fluid Kinematics:
Types of flow – Velocity and acceleration – Local and convective acceleration – Streamline, pathline, streakline – Continuity equation – Rotational flow, circulation – Velocity potential and stream functions.

Flow Through Pipes:
Energy losses – Hydraulic Gradient Line (HGL) and Total Energy Line (TEL) – Equivalent pipe – Pipes in series and parallel – Siphon – Transmission of power.

SECTION B

Fluid Dynamics:
Euler's equation – Bernoulli's equation – Energy equation – Momentum equation – Kinetic energy and momentum correction factors – Flow along curved streamline – Free and forced vortex motion.

Dimensional Analysis and Similitude:
Units and dimensions – Dimensional homogeneity – Rayleigh's method – Buckingham's π theorem – Dimensionless numbers – Geometric, kinematic, and dynamic similarity – Model studies.

Laminar and Turbulent Flow:
Reynolds number – Critical velocity – Laminar flow in pipes – Turbulent flow – Darcy-Weisbach equation – Minor losses – Hydraulic and energy gradient lines.

Flow Measurement:
Manometers – Pitot tube – Venturimeter – Orifice meter – Orifices and mouthpieces – Notches and weirs.

(C) RECOMMENDED BOOKS

1. D.S. Kumar, *Fluid Mechanics and Fluid Power Engineering*, S.K. Kataria & Sons.
2. A.K. Jain, *Fluid Mechanics*, Khanna Publishers.
3. Wylie & Streeter, *Fluid Mechanics*, McGraw Hill.
4. Fox & McDonald, *Introduction to Fluid Mechanics*, John Wiley & Sons.
5. Shames, *Mechanics of Fluids*, McGraw Hill.
6. K. Subramanya, *Theory and Applications of Fluid Mechanics*, Tata McGraw Hill.
7. S.C. Gupta, *Fluid Mechanics & Hydraulic Machines*, Pearson Education.
8. Douglas et al., *Fluid Mechanics*, Pitman.

(D) COURSE OUTCOMES

1. Students will understand different types of fluid flows and methods of describing fluid motion.
2. Students will apply principles of fluid statics to analyze forces on surfaces and floating bodies.
3. Students will understand fluid kinematics and apply continuity equation and flow concepts.
4. Students will apply Bernoulli's equation in solving pipe flow and discharge measurement problems.
5. Students will compute friction losses in laminar and turbulent flows.
6. Students will analyze pipe flow systems and design pipe networks.
7. Students will apply dimensional analysis and similitude in fluid flow problems.

➢ **Course Profile**

Course Title	Fluid Mechanics
Type	Core Course (UG Programme)
Session / Semester	2026–27 / Semester III
Subject Code	**CVLB2303T**
Credits	4.0
L–T–P	3–1–0
Maximum Marks	100
Evaluation Scheme	External (50) + Internal (50)
End-Sem Exam Duration	3 Hours
Minimum Passing Marks (External)	40%

CVLB2304T-SOLID MECHANICS

(Session 2026–27)
SEMESTER III

Subject Code:CVLB2304TCredits:4.0
Time: 3 Hrs**Max. Marks:** 100**Min Pass Marks:** 40%**External:** 50

(A) COURSE OBJECTIVES

At the completion of the course, the students will be able to:

- Understand fundamental concepts of stress, strain, and elastic behavior of materials.
- Analyze simple and complex stress systems using theoretical and graphical methods.
- Determine shear force and bending moment in beams under different loading conditions.
- Evaluate bending and shear stresses in beams and structural members.
- Understand torsion in shafts and analyze combined stress conditions.
- Study stability of columns and behavior under buckling.
- Apply strain energy concepts and failure theories in design.

(B) SYLLABUS

SECTION A

Stresses and Strains:
Concept and types of stress and strain – Stress-strain curves – Generalized Hooke's Law – Young's modulus, Bulk modulus, Modulus of rigidity – Poisson's ratio – Volumetric strain – Bars of varying section – Composite bars – Temperature stresses – Relations between elastic constants.

Complex Stresses:
Normal and shear stresses on inclined planes – Combination of stresses – Pure shear – Principal stresses and planes – Mohr's circle – Principal strains – Computation of principal stresses from principal strains.

Shear Force and Bending Moment:
Types of beams – Supports and loading – Sign conventions – SFD and BMD for simply supported, cantilever, and overhanging beams – Relationship between load, shear force, and bending moment.

Bending and Shear Stresses:
Theory of simple bending – Bending equation – Stresses in beams – Composite and flitched beams – Unsymmetrical bending (intro) – Shear stress distribution across sections.

SECTION B

Torsion:
Torsion of circular shafts – Torsion equation – Power transmission – Combined bending and torsion – Equivalent bending moment and torque.

Columns and Struts:
Euler's buckling theory – End conditions – Limitations – Rankine's formula – Columns with eccentric loading.

Strain Energy:
Strain energy due to axial load, bending, shear, and torsion – Impact loading – Theories of failure.

Dams, Chimneys & Retaining Walls:
Introduction – Analysis of gravity dams – Chimneys and retaining walls – Eccentricity – Core of section – Middle third rule – Wind pressure on chimneys.

(C) RECOMMENDED BOOKS

1. E.P. Popov, *Engineering Mechanics of Solids*, PHI, New Delhi.
2. Timoshenko & Gere, *Mechanics of Materials*, CBS Publishers.
3. Pytel &Kiusalaas, *Mechanics of Materials*, Cengage Learning.
4. Gere, *Mechanics of Materials*, Cengage Learning.
5. D.K. Singh, *Mechanics of Solids*, Pearson Education.
6. Shames & Pitarresi, *Solid Mechanics*, PHI.
7. Sadhu Singh, *Strength of Materials*, Khanna Publishers.
8. S.M.A. Kazimi, *Strength of Materials*.

(D) COURSE OUTCOMES

1. Students will understand basic properties of solids such as stress, strain, elasticity, and deformation.
2. Students will analyze stress and strain relationships in structural elements.
3. Students will determine shear force and bending moment and draw SFD & BMD.
4. Students will evaluate bending and shear stresses in beams.
5. Students will analyze torsion in shafts and combined stress conditions.

➢ **Course Profile**

Course Title	Solid Mechanics
Type	Core Course (UG Programme)
Session / Semester	2026–27 / Semester III
Subject Code	**CVLB2304T**
Credits	4.0
L–T–P	3–1–0
Maximum Marks	100
Evaluation Scheme	External (50) + Internal (50)
End-Sem Exam Duration	3 Hours
Minimum Passing Marks (External)	40%

CVLB2305T-Technical Writing, Report Writing & Presentation Skills

Subject Code:CVLB2305T Credits:3.0
Time: 3 Hrs**Max. Marks:**100**Min Pass Marks:**40**External:** 50

Section A

Fundamentals of Technical Writing- Principles , challenges, modes , Grammar essentials and common errors, Technical Vocabulary, Writing definitions, descriptions and processes

Technical Writing Applications- Lab tests procedures, data interpretation, writing abstracts and summaries

Report Writing- Types (lab report, site report, project report) , structure of the report, Project proposal, drafting and revision,Formal/ Informal letter writing, writing scientific papers (structure and citation)

Section B

Presentation Skills- Structuring of presentations, use of various visual aids and technology, handling Q&A, Team presentations

Professional Communication Skills- Group discussion techniques, Resume/CV writing, Interview skills, workplace communication.

Recommended Books

- Technical Communication: A Practical Approach by William S. Pfeiffer and Kaye E. Adkins
- Technical Report Writing Today by Daniel G. Riordan and Steven E. Pauley, Latest Edition
- Technical Writing, Presentation Skills, and online Communication: Professional Tools and Insight.
- Public Speaking and Technical Writing Skills for Engineering Students, Second Edition
By Lea Nurske.

CVLB2306T- Environment & Road Safety Awareness

(Session 2026–27)
SEMESTER III

Subject Code:CVLB2306T Credits:0
Time: 3 Hrs**Max. Marks:**70 **Min Pass Marks:**28

(A) COURSE OBJECTIVES

At the completion of the course, the students will be able to:

- Understand environmental concepts, ecosystem structure, and biodiversity.
- Analyze natural resources and environmental pollution issues.
- Gain awareness of environmental laws, policies, and sustainability practices.
- Understand human-environment interactions and disaster management.
- Develop awareness regarding road safety rules and first aid.
- Analyze the impact of stubble burning and explore sustainable alternatives.

(B) SYLLABUS

SECTION A

Introduction to Environmental Studies:
Definition, scope, and importance – Multidisciplinary nature – Biosphere components: lithosphere, hydrosphere, atmosphere.

Ecosystem & Biodiversity:
Ecosystem structure and types – Biodiversity: definition, values, threats, and conservation – Levels of biodiversity – Bio-geographical zones of India – Hotspots – India as a mega-biodiversity nation – Endangered and endemic species – Ecosystem services.

Natural Resources:
Renewable and non-renewable resources – Land degradation, soil erosion, desertification – Deforestation and its impacts – Water resources: use, over-exploitation, floods, droughts – Energy resources and alternate energy sources.

Environmental Pollution:
Types, causes, effects, and control of air, water, soil, and noise pollution – Nuclear hazards – Solid waste management – Source segregation – Urban and industrial waste – Case studies.

SECTION B

Environmental Laws & Policies:
Environmental Protection Acts (Air, Water, Wildlife, Forest) – Enforcement issues – Role of individuals – Climate change, global warming, ozone depletion, acid rain.

Human Communities & Environment:
Population growth and impacts – Sanitation and hygiene – Rehabilitation of displaced people – Disaster

management (floods, earthquakes, cyclones, landslides) – Environmental movements (Chipko, Silent Valley, Bishnoi) – Environmental ethics.

Environmental Awareness:
Environmental communication – Public awareness – Case studies (e.g., CNG in Delhi).

Road Safety Awareness:
Traffic rules and signs – Traffic offences and penalties – Driving license process – Role of first aid.

Stubble Burning:
Meaning – Impacts on health and environment – Management techniques – Policies in Punjab.

Field Work:
Visits to environmental sites – Study of flora/fauna – Study of ecosystems (pond, river, etc.).

(C) SUGGESTED BOOKS

1. Carson, R., *Silent Spring.*
2. Gadgil & Guha, *This Fissured Land.*
3. Gleeson & Low, *Global Ethics and Environment.*
4. Gleick, *Water in Crisis.*
5. Groom et al., *Principles of Conservation Biology.*
6. McNeill, *Something New Under the Sun.*
7. Odum, *Fundamentals of Ecology.*
8. Pepper et al., *Environmental and Pollution Sciences.*
9. Rao & Datta, *Waste Water Treatment.*
10. Raven et al., *Environment.*
11. Rosencranz et al., *Environmental Law in India.*
12. Singh et al., *Ecology & Environmental Science.*

(D) COURSE OUTCOMES

1. Students will understand environmental concepts, ecosystems, and biodiversity.
2. Students will analyze natural resources and environmental pollution issues.
3. Students will understand environmental laws and sustainability practices.
4. Students will develop awareness about road safety and first aid.
5. Students will evaluate environmental issues such as stubble burning and propose solutions.

- **Course Profile**

Course Title	Environment & Road Safety Awareness
Type	Audit / Awareness Course
Session / Semester	2026–27 / Semester III
Subject Code	**CVLB2306T**
Credits	0.0
L–T–P	2–0–0

Maximum Marks	100
Evaluation Scheme	Theory (70) + Internal (15) + Field Work (15)
End-Sem Exam Duration	3 Hours
Minimum Passing Marks	40% (Theory)

CVLB2351P-SURVEY-I LAB

L	T	P	Credits
0	0	2	1.0

1 Measurement of distance, ranging a line, plotting of details in chain survey.

2 Measurement of bearing and angles with compass, adjustment of traverse by graphical method.

3 Different methods of leveling, height of instrument, rise & fall methods.

4 Plane table survey, different methods of plotting two point &three point problem.

Course Outcome:

1. Apply advanced surveying techniques in different fields of civil engineering.
2. Select the advanced surveying technique which is best suited for a work
3. Apply total station and EDM in distance measurement and traversing.

CVLB2352P-FLUID MECHANICS LAB

L	T	P	Credits
0	0	2	1.0

1. To determine the Reynolds's number and hence the type of flow.
2. To determine co-efficient of discharge (c_d) for venturimeter and orifice meter & calibrate Rota meter.
3. To determine the co-efficient of discharge (c_d) through different types of notches i.e. Rectangular & V- notch.
4. To verify the Bernoulli's theorem.
5. To determine the losses due to friction in pipes.
6. To determine the coefficient of Pitot tube and plot the velocity profile across the cross section of pipe.
7. To determine the Metacentric height & position of the metacenter with angle of heel for the ship model.
8. To determine the co-efficient of discharge and co-efficient of velocity for Orifice & Mouthpiece.

Course Outcomes:

Upon successful completion of this course, it is expected that students will be able to:

1. Apply dimensional analysis for design of experimental procedures.
2. Calibrate flow measuring devices used in pipes, channels and tanks.
3. Determine fluid and flow properties.
4. Characterize laminar and turbulent flows.

CVLB2353P-SOLID MECHANICS LAB

L	T	P	Credits
0	0	2	1.0

1. To determine Rockwell hardness number of the specimen of steel / soft metal
2. To determine Brinell hardness number of the specimen of steel / soft metal
3. To determine Vicker's hardness number of the specimen of steel / soft metal
4. To determine the modulus of rigidity of a bar on torsion testing machine (destructive test)
5. To determine the impact strength of a specimen on Izod / Charpy impact testing machine
6. To determine the Young's modulus of the material of a beam simply supported at the ends and carrying a concentrated load at the center
7. To determine the Young's modulus of the strip on tensile testing machine.
8. To study the behavior of the material on universal testing machine

Course outcomes: On completion of the course, the students will be able to evaluate Young Modulus, torsional strength, hardness and tensile strength of given specimens and find stiffness of open coiled and closed coiled springs

CVLB2354P-CONCRETE LAB

L	T	P	Credits
0	0	2	1.0

1. To Determine the Compressive Strength of Cement.
2. Design of a concrete mix in accordance with BIS and ACI guidelines.
3. To Determine the Slump, Compaction Factor and Vee-Bee Time of Concrete.
4. Determination of flexural strength of concrete.
5. Determination of split tensile strength of concrete.
6. Determine the modulus of elasticity.
7. Effect of partial replacement of Cement by fly ash on properties of concrete.
8. To determine the Compressive Strength of hardened Concrete by Non-Destructive Test.
   (Rebound hammer and Ultrasonic pulse velocity)

Books/Manuals :

1 Concrete Manual, Dr. M.L. Gambhir, Dhanpat Rai& Sons Delhi.

2 Concrete Lab Manual, TTTI Chandigarh

Course outcome:

1. Learn to determine the compressive strength, flexural strength and split tensile of concrete.
2. To know about the modulus of elasticity and effect of partial replacement of cement by fly ash on properties of concrete.